

MarketGyan: NEPSE-Impact Dataset, Model Training, and RAG Automation Progress

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Project Supervisor Discussion

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Project Goal

Research contribution

MarketGyan introduces a Nepal-specific market-impact benchmark and model workflow for identifying whether news is relevant to NEPSE, mapping it to event types, sectors, symbols, and evidence, and supporting grounded post-market reporting.

- Focus: Nepal share-market language, entities, and event mechanisms.
- Dataset: human-adjudicated bilingual NEPSE-Impact-500.
- Model: XLM-R and FinBERT baselines; Qwen3-8B QLoRA structured adapter.
- Automation: RAG and deterministic market data remain the source of current facts.

Problem Reframing

Initial issue

General business news often looks financial but does not have a defensible NEPSE impact mechanism.

- Product launches
- Commodity retail prices
- Foreign employment stories
- Non-listed company news

Revised task

Learn market relevance, event taxonomy, symbol/sector mapping, and evidence-grounded potential impact.

- Direct NEPSE impact
- Indirect sector or macro impact
- Hard-negative non-market stories

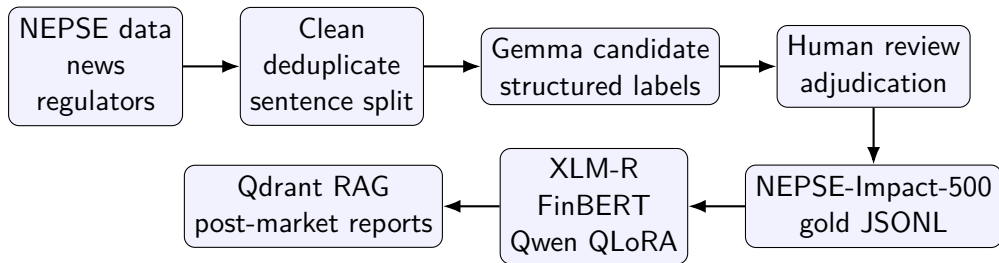
Novelty Position

Reference idea	MarketGyan contribution
FinBERT	Domain adaptation, but for Nepal market relevance rather than generic English finance sentiment.
FinGPT	Data-centric lightweight tuning using a curated, auditable local corpus.
FinLLM survey	Benchmark design, hallucination control, JSON validity, and grounding metrics.
FinTral	Multi-task structured financial reasoning with evidence, sectors, symbols, and impact mechanism.

Claim boundary

LoRA itself is not the novelty. The novelty is the NEPSE-specific ontology, bilingual benchmark, adjudication protocol, and evidence-grounded evaluation.

Data Pipeline



- Numeric market data is deterministic and never model-controlled.
- News/regulatory text is model-assisted but human-adjudicated.

Corpus Status

Component	Count	Status	Notes
Gold records	500	Ready	Human-adjudicated schema-v2 labels.
Direct records	300	Target met	Company or NEPSE-market direct relevance.
Indirect records	100	Target met	Sector or macro mechanism.
Hard negatives	100	Target met	General business but no NEPSE mechanism.
English records	299	Target met	Bilingual benchmark.
Nepali records	200	Target met	Minimum language target met.
Symbol-level records	273	Target met	Needed for listed-company mapping.

Ontology and Label Schema

Core fields

- Relevance: direct, indirect, not relevant
- Event type
- Impact scope
- Impact direction
- Impact horizon
- Impact mechanism

Grounding fields

- NEPSE sectors
- Listed symbols
- Confidence band
- Evidence sentence IDs
- Source URL and date

Important rule

A record is positive only when there is a concrete NEPSE company, sector, market, regulatory, liquidity, or macro mechanism.

Baseline Training Results

Model/task	Accuracy	Macro-F1	Main finding
XLM-R relevance	0.707	0.665	Best reliable relevance baseline; indirect class remains weak.
XLM-R direction	0.634	0.401	Bearish and uncertain are usable; bullish and neutral need work.
FinBERT English direction	0.929	0.636	Strong but only 14 English test rows, so it is a motivation baseline.
Qwen QLoRA first run	0.213	0.250	Failed first iteration: 41.3% JSON validity and all-direct bias.

Interpretation

XLM-R is currently the reliable baseline. Qwen requires compact-output retraining before its result is meaningful.

Error Audit Findings

Baseline errors

- XLM-R relevance: 22/75 errors.
- XLM-R direction: 15/41 errors.
- FinBERT: 1/14 errors.

Qwen errors

- 44/75 invalid or truncated JSON outputs.
- 10 hard-negative false positives among valid JSON.
- Strong bias toward direct relevance.

Manual revalidation target

Prioritize hard negatives that both XLM-R and Qwen mark as relevant, plus bullish direct records predicted as uncertain.

Why Qwen Failed in First Run

- Full gold JSON targets were too verbose for stable generation.
- Qwen was asked to generate summaries, rationales, evidence text, tags, and labels in one output.
- The chronological split produced a language/source shift: train was mostly English/direct while test had many Nepali hard negatives.
- The model learned output style but not a strong not-relevant gate.

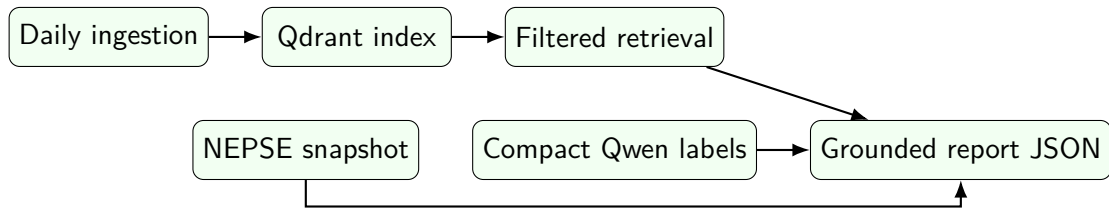
Fix already implemented

Balanced duplicate-grouped split plus compact Qwen labels: canonical labels, sectors, symbols, confidence, and evidence sentence IDs only.

New Training Scheme

Part	Decision
Split	Balanced duplicate-grouped 70/15/15; chronological retained as robustness check.
Qwen target	Compact JSON only; no summary, rationale, evidence text, tags, or prose.
Hard negatives	Oversample not-relevant, indirect, and Nepali hard-negative examples.
Evaluation	JSON validity, grounding, relevance macro-F1, event macro-F1, direction macro-F1, sector/symbol F1.
Report generation	Use RAG and deterministic data for current facts and citations.

RAG and Automation Design



Key boundary

The adapter classifies and structures evidence. RAG supplies current facts and citations for post-market reports.

Demonstration Path

- 1 Show the NEPSE-Impact-500 corpus and ontology.
- 2 Show human-adjudicated labels and evidence sentence IDs.
- 3 Show XLM-R/FinBERT baseline metrics.
- 4 Show first Qwen failure analysis and the corrective compact-label design.
- 5 Rerun compact Qwen and compare against XLM-R on identical balanced test IDs.
- 6 Use RAG to generate a cited post-market report from retrieved evidence and deterministic market data.

Next Milestones

- ① Revalidate priority error samples from the audit CSV.
- ② Rerun XLM-R and FinBERT on the balanced split.
- ③ Rerun Qwen zero-shot, three-shot, and compact QLoRA.
- ④ Compare Qwen with XLM-R and base Qwen.
- ⑤ Complete second-review agreement on 110 stratified records.
- ⑥ Populate Qdrant and evaluate 30 retrieval queries.
- ⑦ Demonstrate one post-market report with citations and no investment advice.

Summary

What is complete

Data pipeline, schema-v2 ontology, 500 adjudicated gold records, balanced split, baseline training outputs, Qwen failure audit, compact-Qwen fix, and RAG service boundary.

What remains

Revalidate difficult samples, rerun compact Qwen, complete second-review agreement, evaluate retrieval, and demonstrate one grounded post-market report.

Core message

MarketGyan is a Nepal-specific evidence-grounded market-impact benchmark and modeling workflow, not a trading signal or price-prediction system.